Nandhana Nambi
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Education

MS in Health Data Science (2024 -2026), University of Michigan, Ann Arbor, USA

B.Tech. Biotechnology with specialization in Regenerative Medicine (2020 - 2024), SRM Institute of Science and Technology, Chennai, India

Coursework

Computing for Data Analytics (Python, R, C++) | Computing with Big Data | Machine Learning for Health Science (Ongoing) | Biostatistical Inference | Generalized Linear Models | Clinical Trials

Publications and Awards

- Co-author on 3 peer-reviewed publications in Current Vascular Pharmacology (2024), Human Immunology (2025), and Cytokine (2025).
- Best Poster Award - ICB '24 (NEK7-mediated NLRP1 activation in GDM)

Certification

- Oxford Machine Learning Summer School - ML x Bio
- Workshop on Accessing Health Data at University of Michigan
- Google Data Analytics Professional Certificate

Other Interests

Peer Mentor - Biostatistics | Planet Blue Ambassador | MSSISS Committee Organizer (University of Michigan)

Summary

Health Data Science Masters student with experience analyzing clinical and real-world healthcare datasets using Python, R and SQL. Skilled in biostatistics, clinical trial simulation, and safety & market analysis, seeking entry-level data scientist / data analyst roles in Life Sciences / Healthcare industry.

Work Experience

- **Systems Biology Lab, University of Michigan**, Student Researcher, Ann Arbor, MI, U.S - Analyzed high-dimensional metabolic datasets across cellular states using Python/R and computational modeling to identify condition-specific flux patterns. (Ongoing)
- **STATCOM - Data Analyst (Volunteer), Ann Arbor,** MI, U.S - Performed data preprocessing, EDA, and regression in Python/R on real-world datasets for community organizations; delivered visual summaries to support program decisions. (Ongoing)
- **Agilisium Consulting, Westlake Village**, CA, U.S (Remote) - Built CNN-based models to classify histopathology images across three cancer types, achieving ~83% validation accuracy using transfer learning and Grad-CAM. (June 2025 – August 2025)
- **SRM IST Cell Signaling Laboratory and SRM Hospital, Chennai**, India - Managed and analyzed inflammatory marker data from GDM studies in a clinical research setting. (Jan 2023 - July 2024)

Projects

- **Drug Safety & Market Analysis** - Built a PostgreSQL database from FDA FAERS real-world evidence, integrating patient demographics and reporting metadata to support safety analysis and market trends.
- **Breast Cancer Type Prediction** using Wisconsin Diagnostic Breast Cancer Dataset - Built and evaluated classification models (penalized regression, random forest) in Python/R to predict malignant vs benign cases using clinically relevant metrics (AUC, sensitivity, specificity).
- **Phase I dose escalation for various metastatic cancers** - Simulated Phase 1 dose-escalation designs in R to compare toxicity rates and recommended dose levels across strategies, supporting trial design decisions.
- **Integrative ODE and Network-Based Modeling** of HPA Axis Stress Regulation - Developed ODE-based and network-based models of the HPA axis to study stress-response dynamics and identify conditions leading to dysregulated cortisol levels

Skills

- **Programming & Systems -** Python, R, SQL, C, C++, Unix
- **Databases and Data Analysis -** PostgreSQL, Pandas, Scikit-learn
- **Machine Learning -** CNNs, Transfer Learning, Model Evaluation, Grad-CAM
- **Tools & Platforms -** Git, Jupyter Notebook, BigQuery, Linux, Looker
- **Modeling & Simulation -** ODE-based dynamical systems, Berkeley Madonna
- **Biological Software**: SnapGene